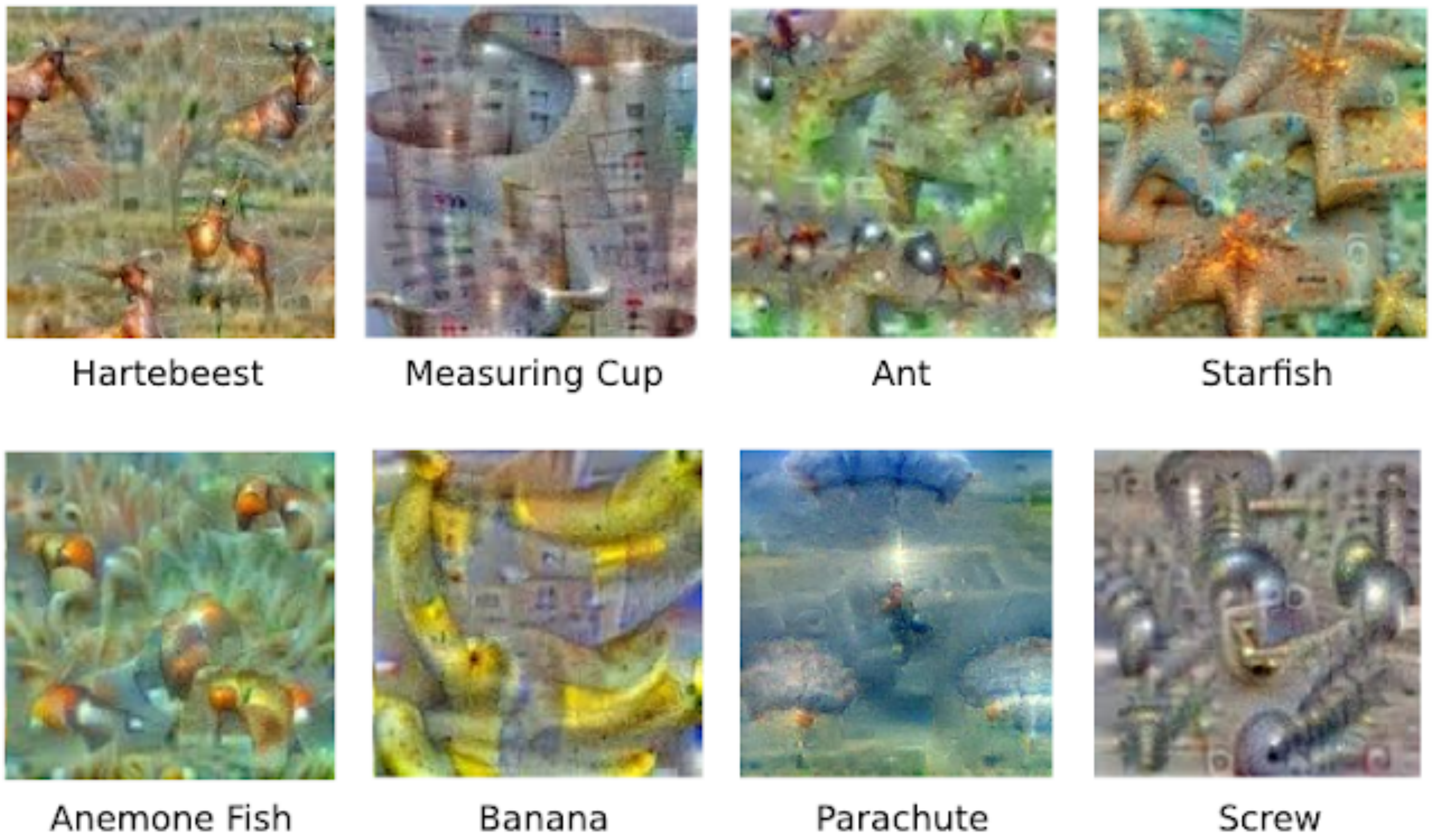
Explaining CNN Outputs

CSC 487: Deep Learning

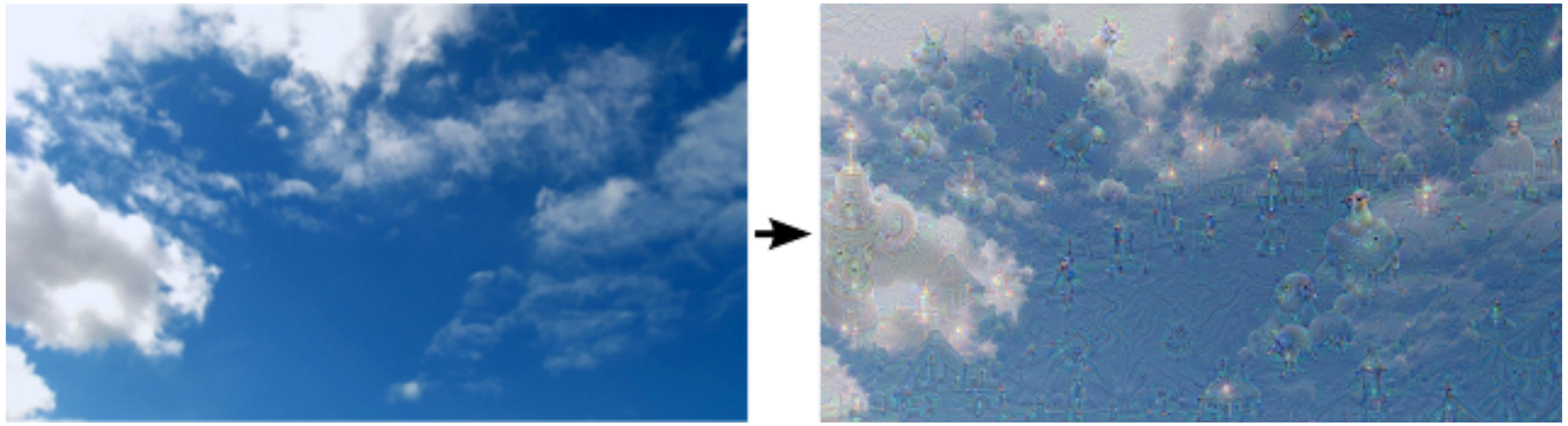
Instructor: Jonathan Ventura

Deep Dream

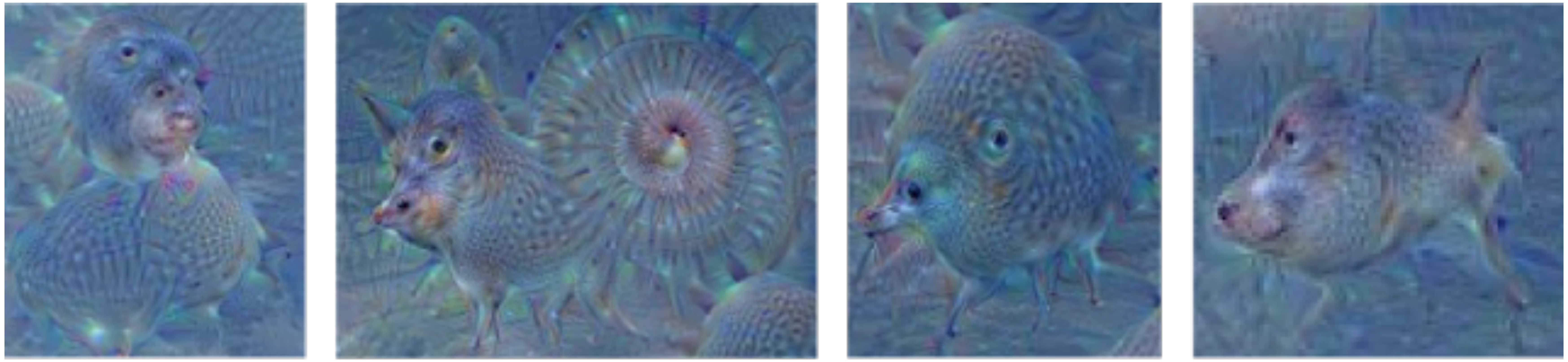


Mordvintesev et al. Inceptionism: Going Deeper into Neural Networks. 2015

Deep Dream

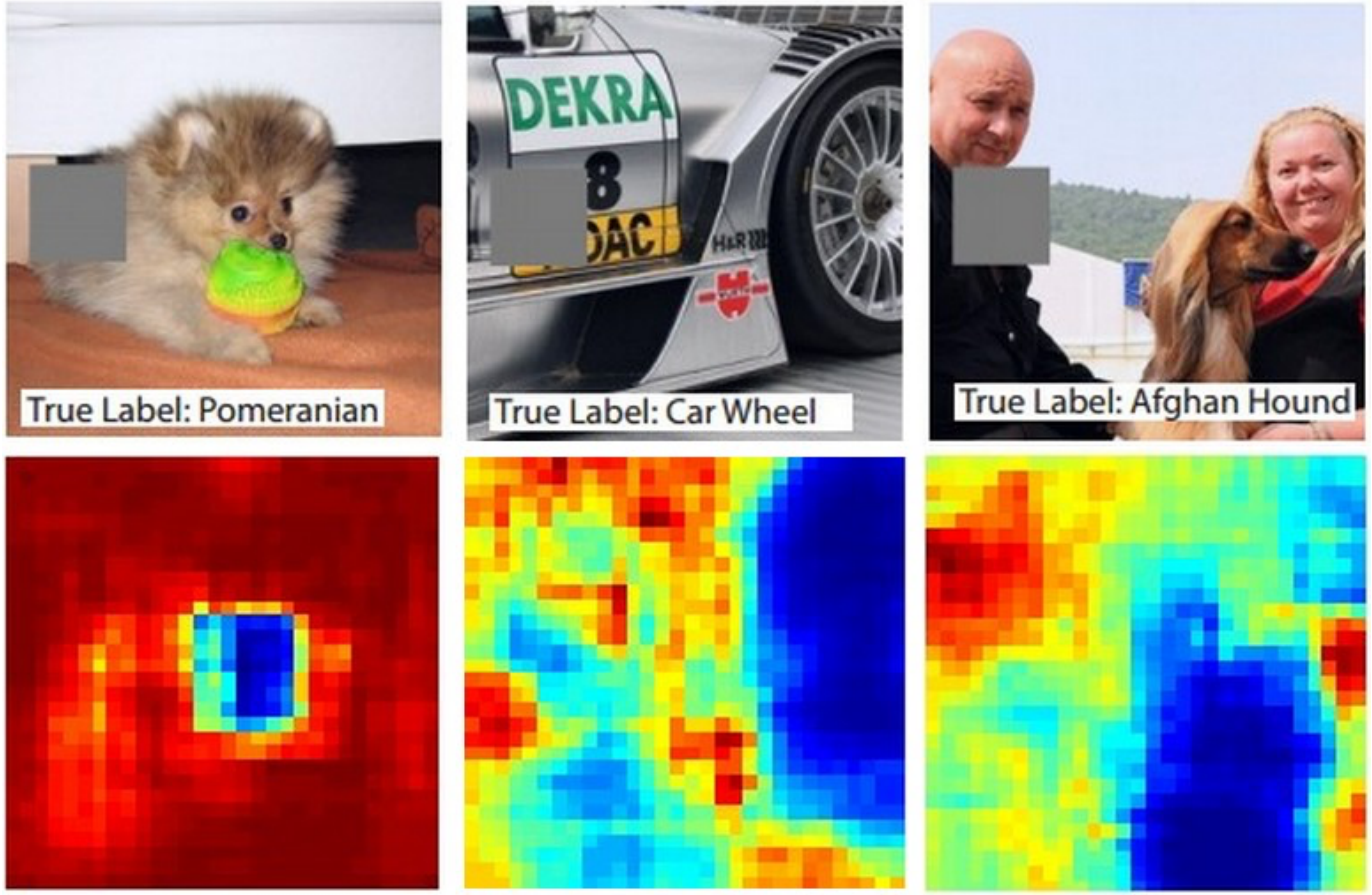


“Whatever you see there, I want more of it!”



"Admiral Dog!" "The Pig-Snail" "The Camel-Bird" "The Dog-Fish"

Visualizing and understanding CNNs

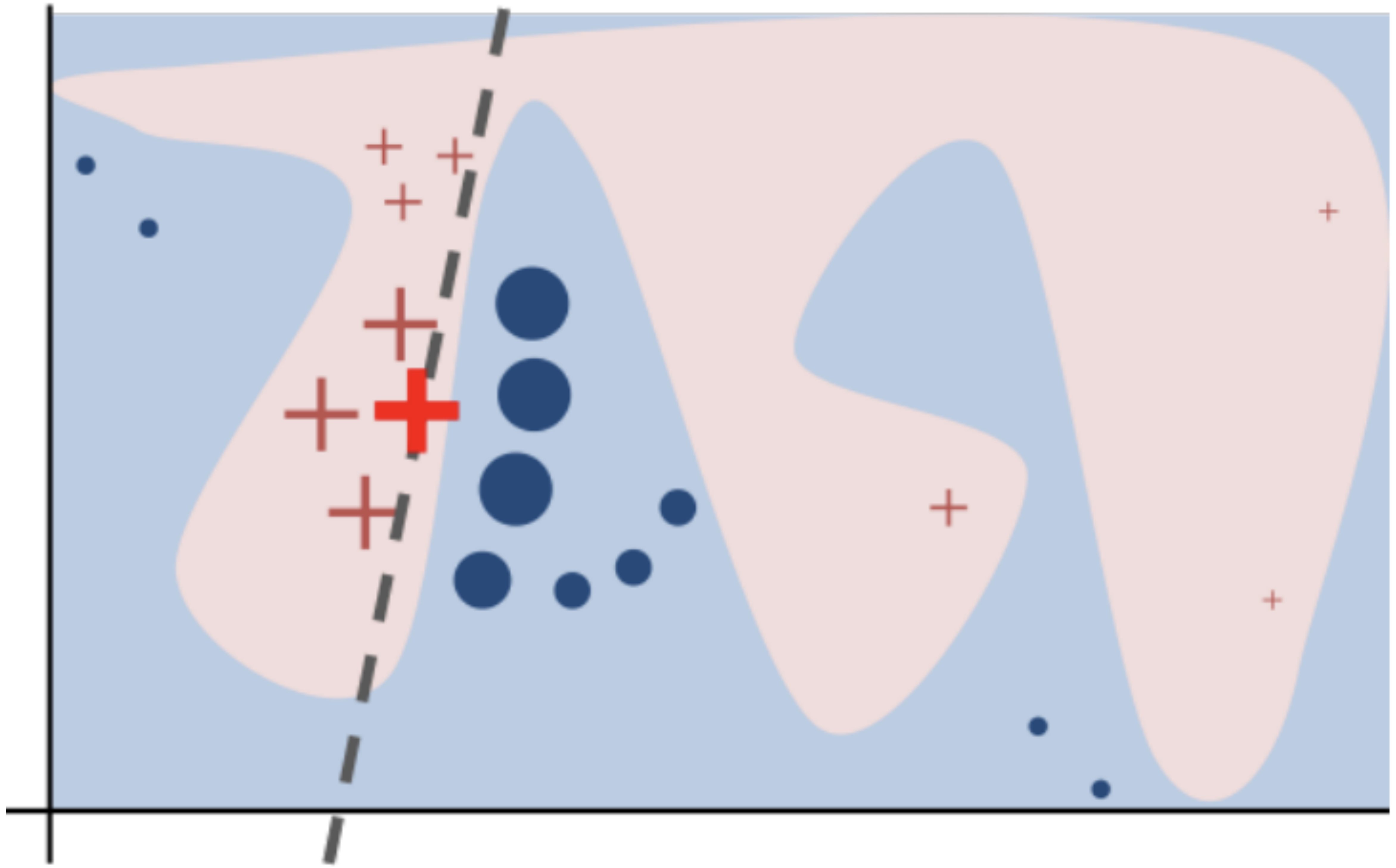


Zeiler, Matthew D., and Rob Fergus. "Visualizing and understanding convolutional networks." *European Conference on Computer Vision*. Springer International Publishing, 2014.

LIME: Local Interpretable Model-Agnostic Explanations

Goal: Explain a decision by an arbitrary ML model

Idea: Approximate the local decision boundary by fitting a linear model to points near the instance being explained.

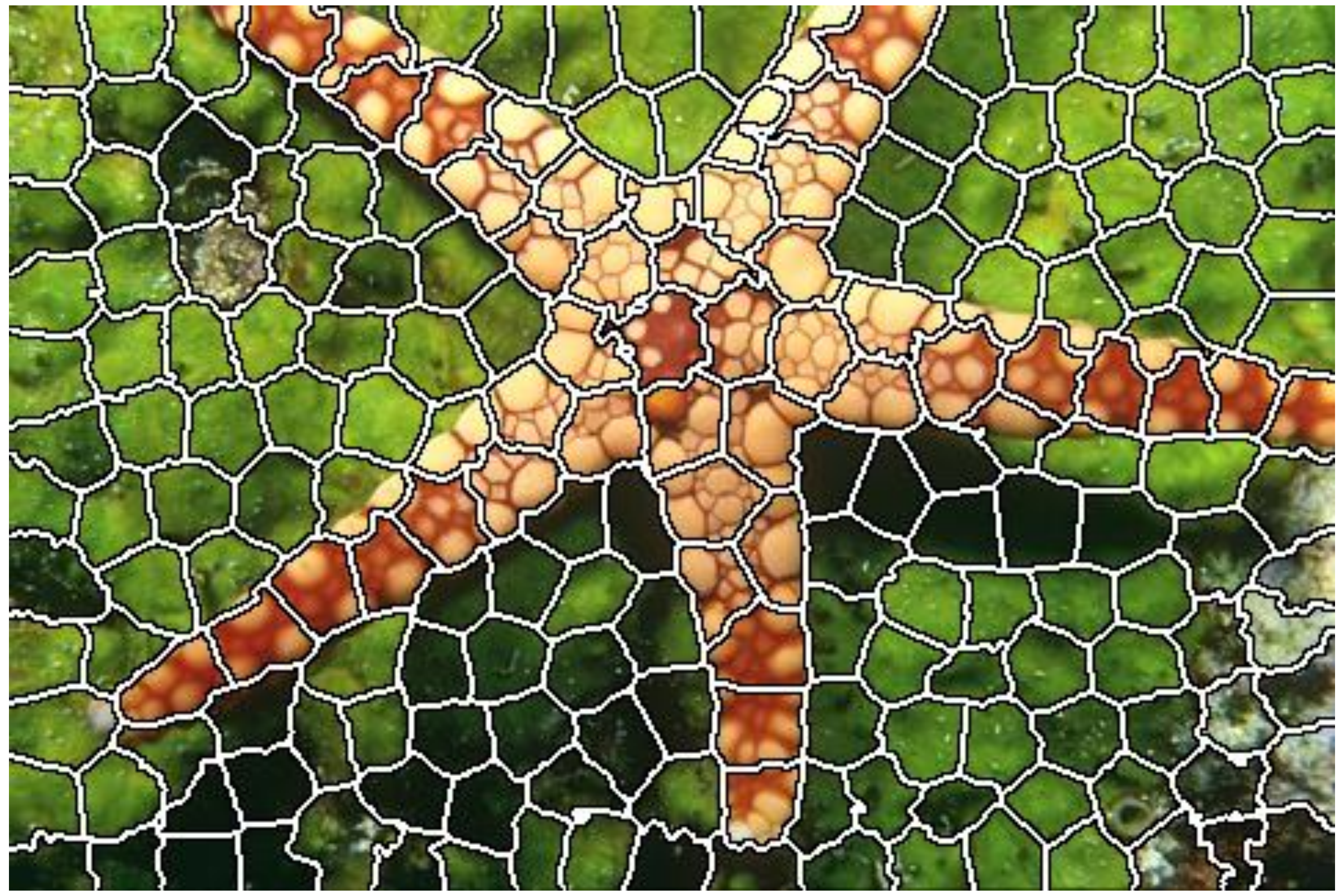


Ribeiro, M. T., Singh, S., & Guestrin, C. (2016, August). " Why should i trust you?" Explaining the predictions of any classifier. In Proceedings of the 22nd ACM SIGKDD international conference on knowledge discovery and data mining (pp. 1135-1144).

Selecting samples

LIME selects samples using an “interpretable representation” of the input, rather than the raw input.

Group pixels into “super pixels” (clusters of similar pixels)
Make a binary vector indicate whether each superpixel is masked out



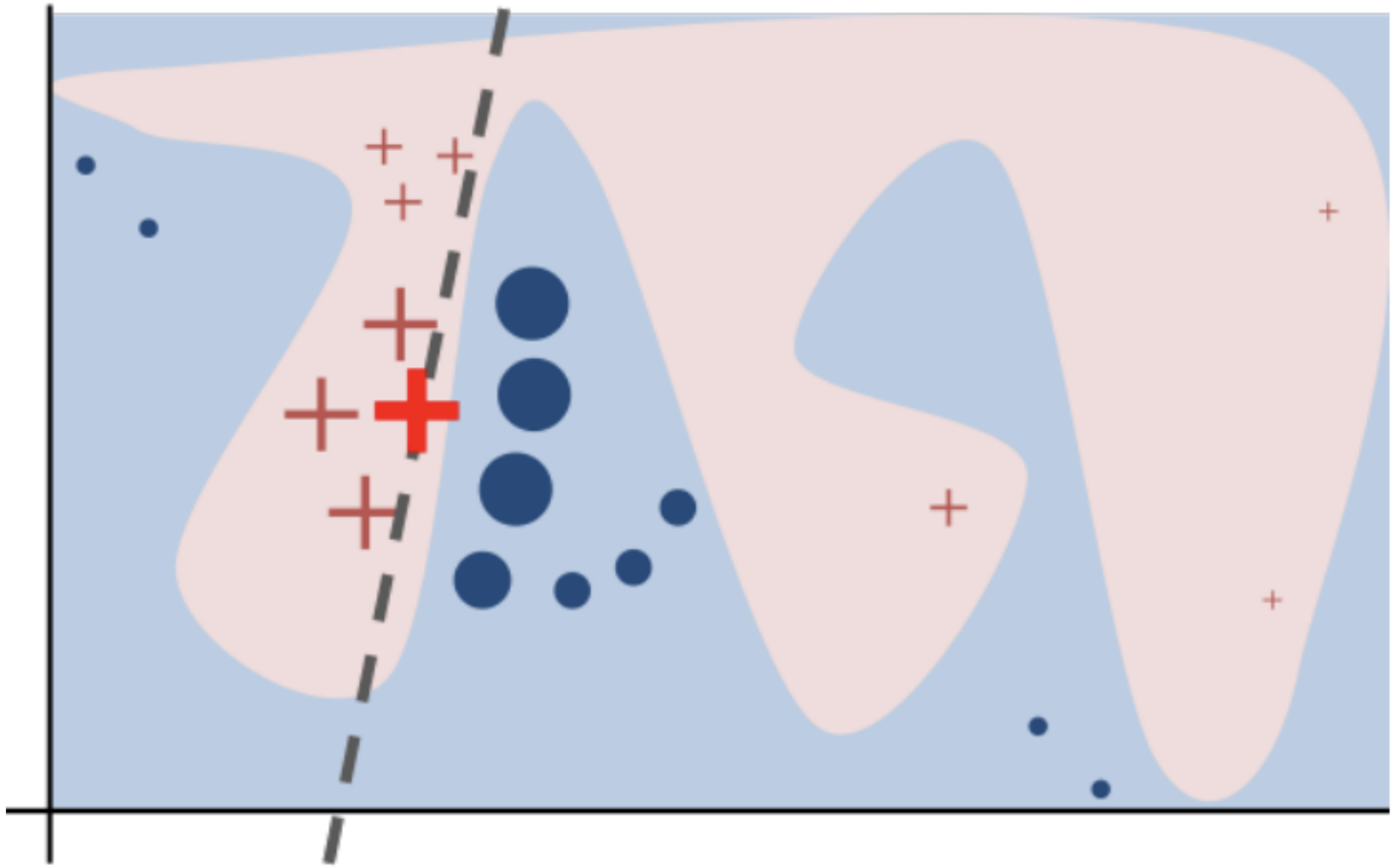
Fitting the linear model

Each randomly masked sample is passed to the ML model to get the model's predicted label

Then we fit a linear model to the set of samples and predicted labels

Binary variables with high coefficients are important to the ML models' decision

Use feature selection method to determine most important superpixels



Example: Image Classification

(a) Original Image (b) Explaining *Electric guitar* (c) Explaining *Acoustic guitar* (d) Explaining *Labrador*

Inception neural network for image classification

Summary

LIME fits a linear model to local decision boundary of an arbitrary ML model

Can help find defects in a dataset or model

Or, can help inspire confidence in a model

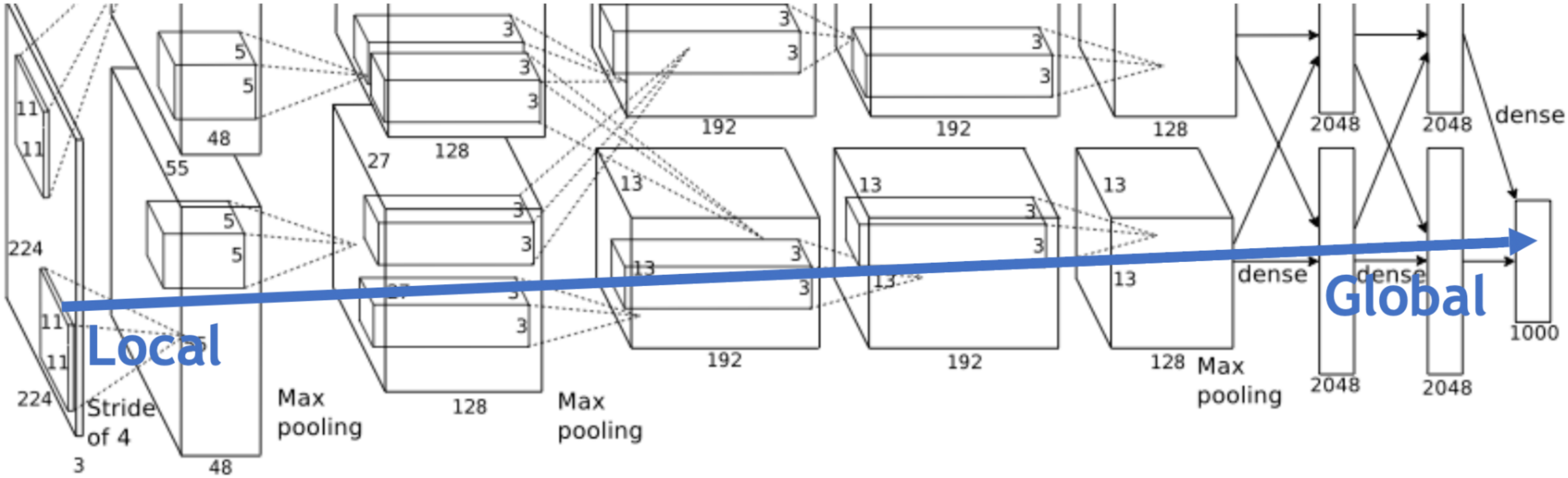
Limitations:

Local model: interprets a single prediction at a time

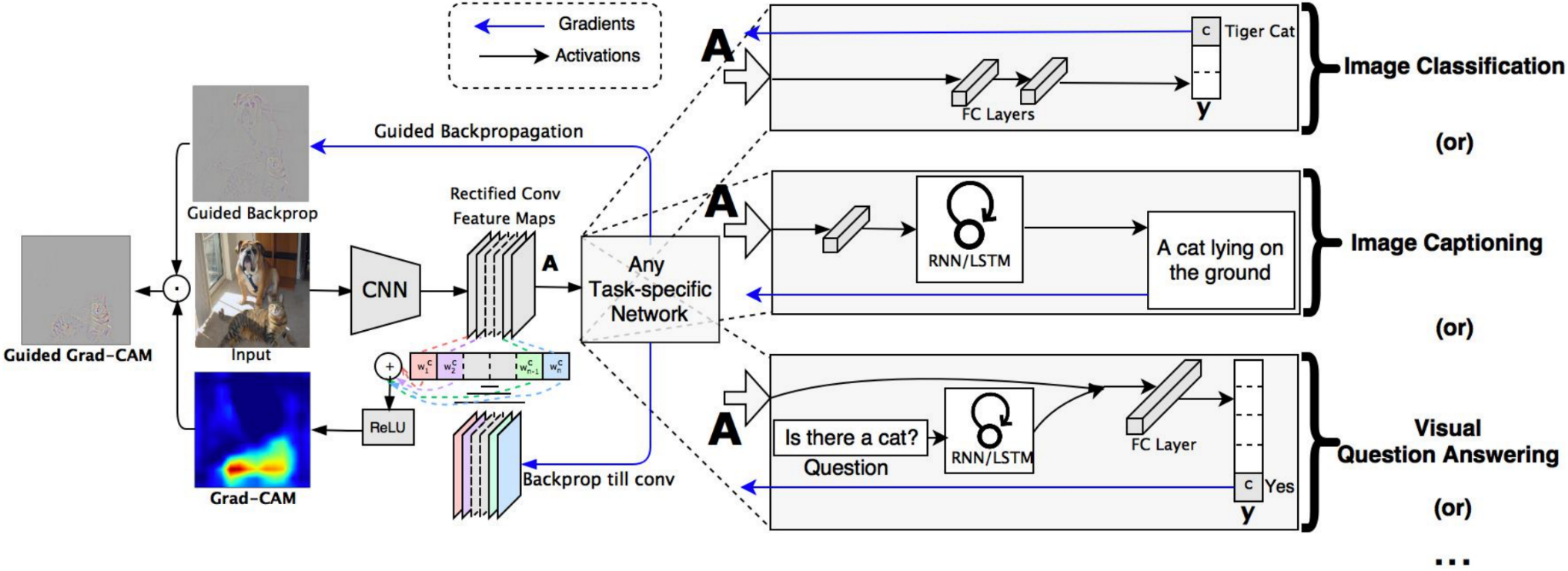
Not a **global** model, to explain the behavior of the model on any input

Convolutional Neural Networks

Krizhevsky, Alex, Ilya Sutskever, and Geoffrey E. Hinton. "ImageNet Classification with Deep Convolutional Neural Networks." *Advances in Neural Information Processing Systems 25* (2012): 1–9. Web.



Grad-CAM: Visual Explanations from Deep Networks



Selvaraju, R. R., Cogswell, M., Das, A., Vedantam, R., Parikh, D., & Batra, D. (2017). Grad-cam: Visual explanations from deep networks via gradient-based localization. In *Proceedings of the IEEE International Conference on Computer Vision* (pp. 618-626).

Computation of neuron importance weight

$$\alpha_k^c = \overbrace{\frac{1}{Z}\sum_i \sum_j}^{\text{global average pooling}} \underbrace{\frac{\partial y^c}{\partial A_{ij}^k}}_{\text{gradients via backprop}}$$

y^c is class being explained

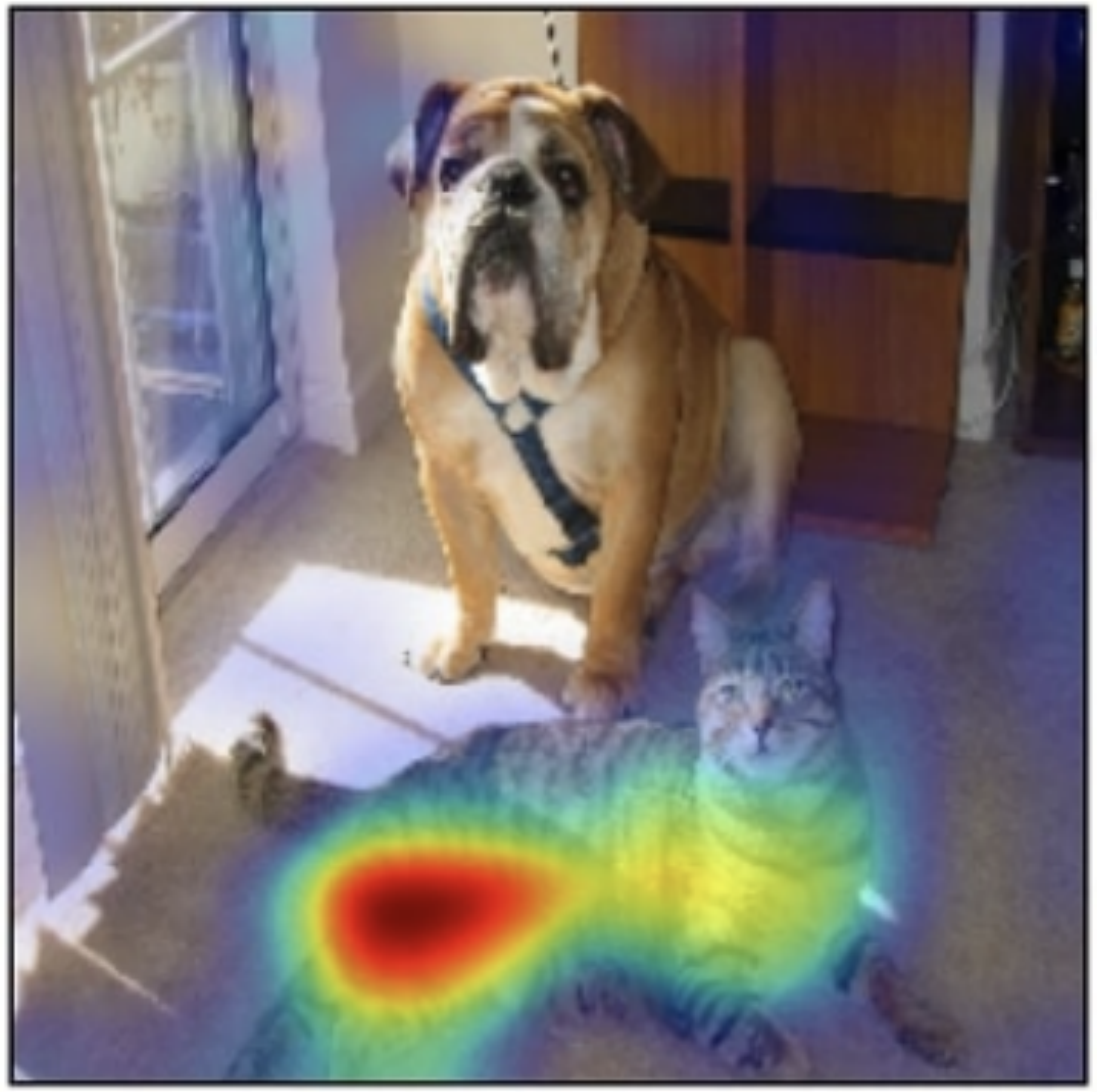
A^k is k-th feature map in layer under consideration

α_k^c is neuron importance weight for feature k

Computation of Grad-CAM Visualization

$$L^c_{\text{Grad-CAM}} = ReLU \underbrace{\left(\sum_k \alpha^c_k A^k \right)}_{\text{linear combination}}$$

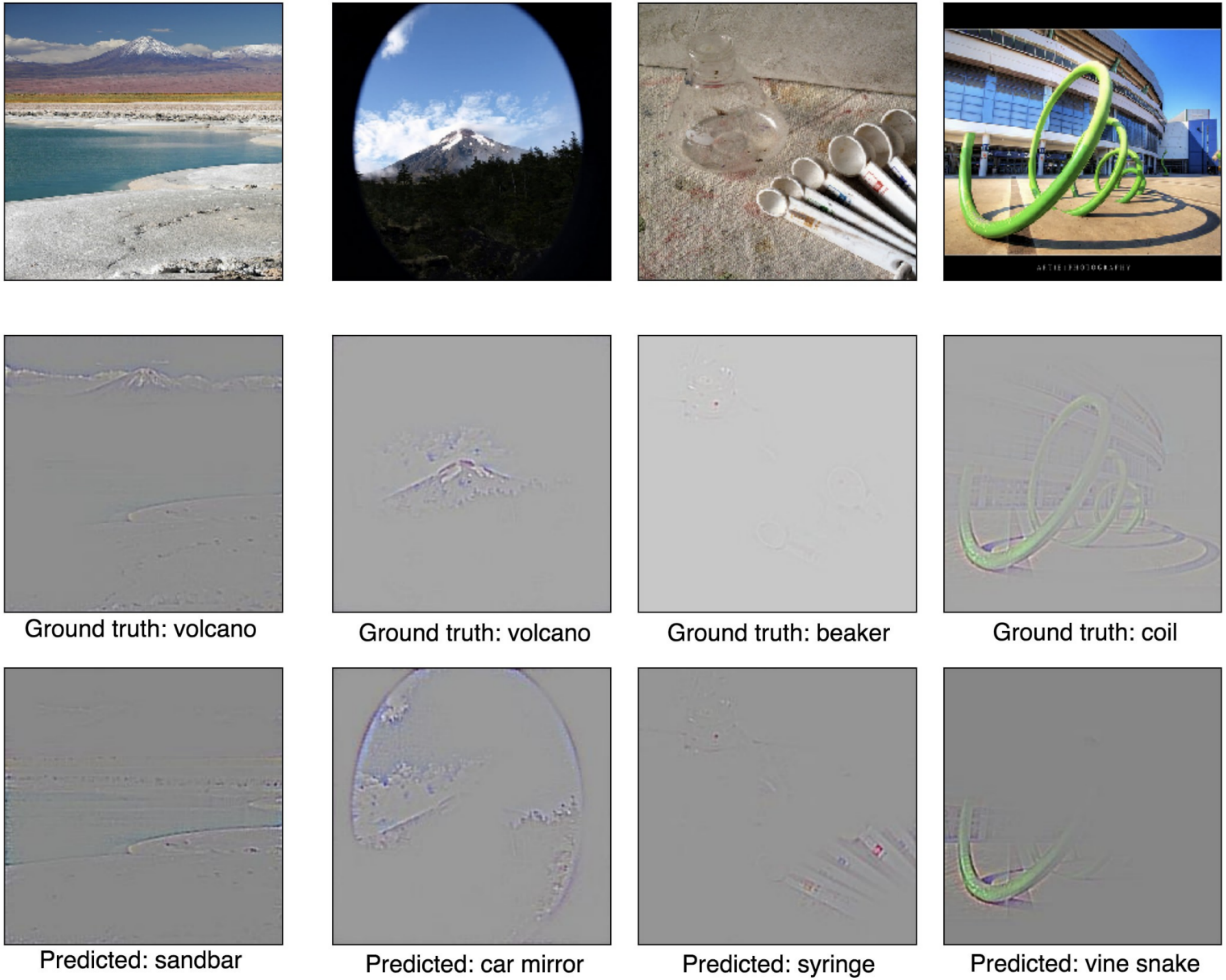
Grad-CAM Examples



(c) Grad-CAM 'Cat'

(i) Grad-CAM 'Dog'

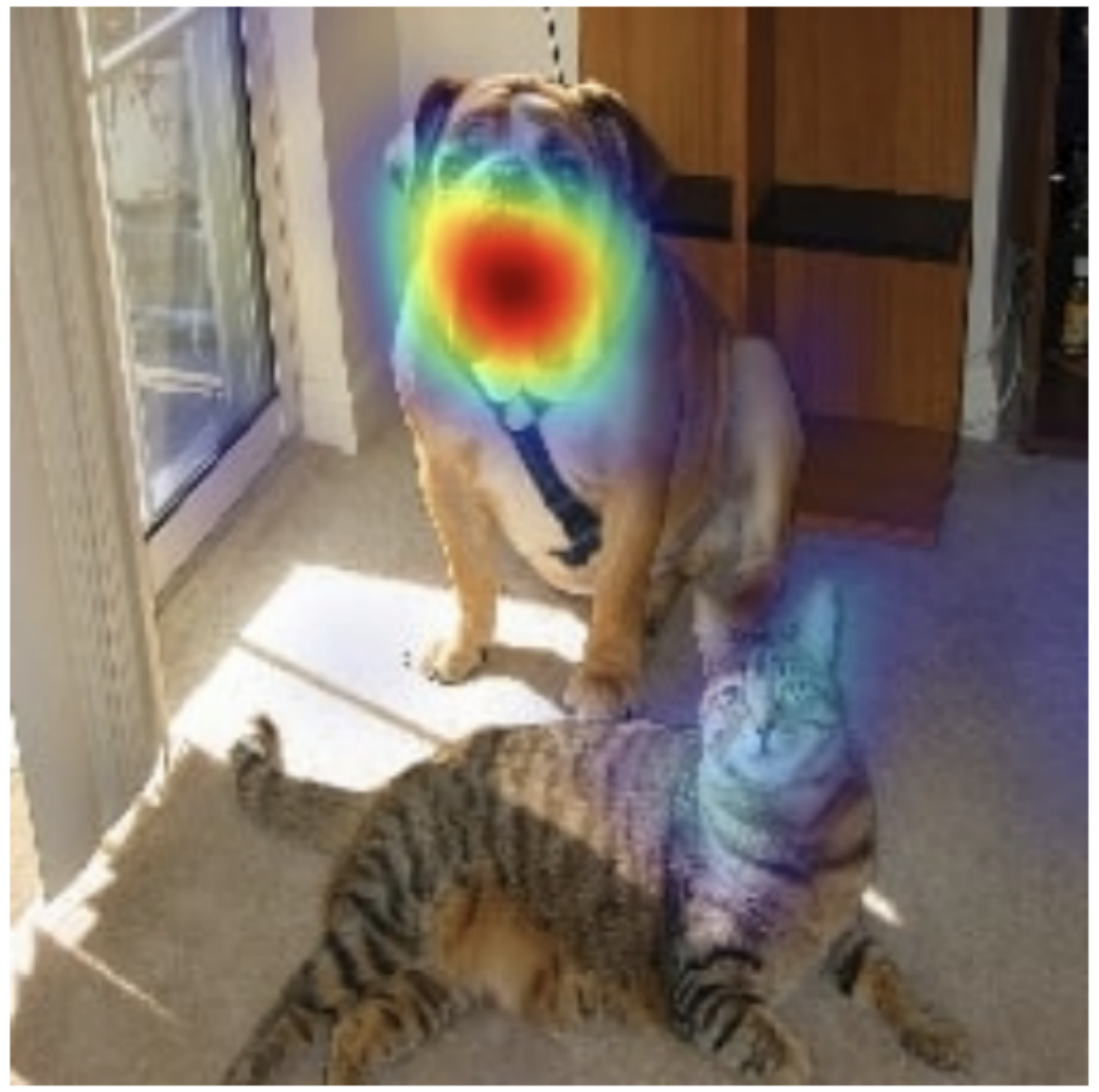
Analyzing failure modes in VGG-16



Analyzing classification of adversarial images

Boxer: 0.4 Cat: 0.2
(a) Original image

Airliner: 0.9999
(b) Adversarial image



Boxer: 1.1e-20
(c) Grad-CAM “Dog”

Tiger Cat: 6.5e-17
(d) Grad-CAM “Cat”

Airliner: 0.9999
(e) Grad-CAM “Airliner”

Space shuttle: 1e-5
(f) Grad-CAM “Space Shuttle”

Analysis of model bias